

Theorem (Chebyshev's sum inequality). *Let $n \in \mathbb{N}$ and let*

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. Then

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \quad (1)$$

Equality holds iff one of the following occurs

1. $n = 1$;
2. the sequence (a_i) is constant, i.e. $a_1 = a_2 = \cdots = a_n$;
3. the sequence (b_i) is constant, i.e. $b_1 = b_2 = \cdots = b_n$.

Proof. Trivial cases. If $n = 1$ both sides equal $a_1 b_1$, so the inequality is an equality. If all a_i are equal (say $a_i \equiv A$), then

$$n \sum_{i=1}^n a_i b_i = nA \sum_{i=1}^n b_i = \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right),$$

and similarly when all b_i are equal. Henceforth assume $n \geq 2$ and that neither (a_i) nor (b_i) is constant.

1. A non-negative double sum. Because the sequences are non-decreasing, for any indices $i < j$ we have

$$a_i - a_j \leq 0, \quad b_i - b_j \leq 0,$$

and therefore

$$(a_i - a_j)(b_i - b_j) \geq 0 \quad (1 \leq i < j \leq n). \quad (1)$$

Summing (??) over all unordered pairs (i, j) with $i < j$ yields a non-negative quantity

$$S := \sum_{1 \leq i < j \leq n} (a_i - a_j)(b_i - b_j) \geq 0. \quad (2)$$

2. Expanding the double sum. For each pair (i, j) ,

$$(a_i - a_j)(b_i - b_j) = a_i b_i + a_j b_j - a_i b_j - a_j b_i.$$

Hence

$$S = \sum_{i < j} (a_i b_i + a_j b_j - a_i b_j - a_j b_i) \quad (2)$$

$$= \underbrace{\sum_{i < j} a_i b_i}_A + \underbrace{\sum_{i < j} a_j b_j}_B - \underbrace{\sum_{i < j} a_i b_j}_C - \underbrace{\sum_{i < j} a_j b_i}_D. \quad (3)$$

Counting the diagonal terms. Each product $a_k b_k$ appears once for every index that can be paired with a different index, i.e. $n - 1$ times. Consequently

$$A + B = (n - 1) \sum_{k=1}^n a_k b_k. \quad (4)$$

Counting the mixed terms. Every mixed product $a_i b_j$ with $i \neq j$ occurs exactly once, either in C (when $i < j$) or in D (when $j < i$). Therefore

$$C + D = \sum_{\substack{i,j=1 \\ i \neq j}}^n a_i b_j = \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right) - \sum_{k=1}^n a_k b_k. \quad (5)$$

Substituting (??) and (??) into (??) gives

$$S = (n - 1) \sum_{k=1}^n a_k b_k - \left[\left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right) - \sum_{k=1}^n a_k b_k \right] \quad (3)$$

$$= n \sum_{k=1}^n a_k b_k - \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right). \quad (6)$$

3. Chebyshev's inequality. From (??) we have $S \geq 0$. Using the identity (??),

$$0 \leq S = n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right),$$

which is exactly the claimed inequality.

4. Equality condition. Equality $S = 0$ in (??) occurs precisely when each term $(a_i - a_j)(b_i - b_j)$ vanishes. For a fixed pair $i < j$,

$$(a_i - a_j)(b_i - b_j) = 0 \iff a_i = a_j \text{ or } b_i = b_j.$$

If there exists a pair with $a_i \neq a_j$, then we must have $b_i = b_j$ for every such pair, forcing all b_k to be equal. Conversely, if some pair satisfies $b_i \neq b_j$, then all a_k must be equal. Thus equality holds exactly in one of the mutually exclusive situations listed in the theorem statement: $n = 1$, or the a -sequence is constant, or the b -sequence is constant. □